Enhanced Dimensionality Reduction for Open-Source Face Image Quality Assessment via Weighted and Variance-Adaptive Principal Component Analysis

By Edward Mendoza

ABSTRACT: Facial recognition has become a cornerstone of modern biometric identification, with applications ranging from border control to surveillance and consumer verification. The quality of the facial images employed in these systems has a direct impact on identification accuracy, thus rigorous and computationally efficient quality assessment is necessary. Previous research demonstrated that Principal Component Analysis (PCA) can compress Open-Source Face Image Quality (OFIQ) measures across brightness, occlusion, and margin groups, resulting in a 20% dimensionality reduction while maintaining around 78.3% data fidelity using a single principal component. However, relying just on PC1 leaves residual variation unaccounted for, which limits reconstruction accuracy, particularly in categories such as occlusion. This study suggests three changes to the current pipeline: extending compression to include the second principal component (PC2), introducing variance-proportional weighting of PC1 and PC2 so that each component contributes proportionally to its explanatory power, and replacing sequential pre- and post-PCA scaling with an integrated global scaling applied uniformly during transformation. To assess generalizability across varied real-world situations, testing is expanded beyond the Labeled Faces in the Wild dataset to include VGGFace2, Flickr Faces HQ, and CelebA, which provide more variance in race, age, and capturing environment. Following the defined validation approach, test photos will be subjected to fake brightness and margin adjustments to replicate unsatisfactory capture situations. Together, our advances aim to significantly improve the original framework's accuracy and computing efficiency, making PCA-based OFIQ compression a more realistic and practical approach for large-scale biometric quality evaluation.

I. INTRODUCTION

Among the various modalities employed in modern biometric systems, facial recognition has emerged as the most pervasive, owing to its passive acquisition, scalability, and compatibility with existing surveillance and consumer infrastructure. Over the past two decades, its integration into personal, commercial, security, and governmental applications has expanded considerably. Facial recognition, as defined for the purposes of this paper, refers to the capacity of a computational system to identify or verify an individual's identity through the analysis of distinctive facial features extracted from images or video. One of the earliest large-scale deployments of this technology occurred at the 2001 Super Bowl in Tampa, Florida, where facial recognition software developed by Viisage Technology was used to screen over 100,000 attendees against a criminal mugshot database — an effort that, despite yielding 19 identifications, revealed significant limitations including a 43% false rejection rate and degraded performance under variable lighting and off-angle poses [1]. While far removed from contemporary capabilities, this deployment illustrates the longstanding recognition of facial biometrics as a practical identification tool and the persistent challenges that have driven subsequent research.

In contemporary deployments, facial recognition systems operate within broader multi-modal biometric architectures, necessitating the development of interoperability standards to ensure consistent and reliable performance across heterogeneous platforms. The most significant of these is ISO/IEC 29794-5, which establishes standardized quality metrics for facial imagery with the explicit aim of ensuring uniformity and alignment across biometric systems industry-wide [2]. Within this framework, attributes such as luminosity, occlusion, and sharpness are recognized as principal determinants of facial image quality. This is of considerable practical importance, as the presence of low-quality imagery in training or inference pipelines has been shown to directly degrade recognition accuracy and introduce systemic bias. In response, open-source quality assessment frameworks such as the Open-Source Face Image Quality (OFIQ) system were developed. First released in 2020, OFIQ serves as the official reference implementation of ISO/IEC 29794-5 and provides standardized scalar quality metrics to support facial image curation and

quality-gated acquisition in both US and EU biometric contexts [3]. Despite its utility, OFIQ's comprehensive metric output introduces substantial computational overhead when applied to large-scale datasets, posing challenges for real-time and resource-constrained deployments.

A prior study titled "Redundancy Reduction and Dimensionality Reduction using Principal Component Analysis for Open-Source Face Image Quality Assessment" proposed a dimensionality reduction approach to address this limitation [4]. That work applied Principal Component Analysis (PCA) to OFIQ scalar outputs, grouping correlated metrics into three composite categories — brightness, occlusion, and facial margins — and extracting a single principal component from each to serve as a representative quality indicator. For instance, Dynamic Range, Luminance Mean, and Luminance Variance were consolidated into a single brightness composite, reducing metric redundancy while theoretically preserving the informational content of the original measures. This approach achieved a 20% reduction in dimensionality, from 29 to 23 metrics, while retaining approximately 78% reconstruction agreement with the original OFIQ outputs. However, several limitations were identified. The pipeline relied on multiple disconnected tools rather than a unified processing framework, validation was conducted exclusively on the Labeled Faces in the Wild (LFW) dataset which constrained the generalizability of findings, and the 22% loss in reconstruction fidelity — attributable in part to the use of a single principal component and inconsistent pre- and post-PCA scaling — represents a meaningful accuracy tradeoff that limits the practical viability of the approach.

The present study seeks to address these limitations and advance upon the prior framework in two substantive directions. First, a modified PCA formulation is proposed that incorporates a second principal component (PC2) alongside PC1, applies variance-proportional weighting to each component's contribution to the composite score, and replaces the prior sequential scaling operations with an integrated global scaling step applied uniformly during transformation. These modifications are intended to improve reconstruction fidelity while preserving the computational efficiency gains achieved through dimensionality reduction. Second, the study expands the scope of dataset validation to assess the generalizability of the modified framework across diverse imaging conditions. In addition to LFW, training and validation are conducted using CelebA, VGGFace2, and Flickr Faces HQ — datasets that collectively encompass a broader range of identities, acquisition environments, resolutions, and demographic characteristics than any single dataset affords.

Section II reviews related work pertaining to PCA enhancement techniques and the influence of dataset diversity on model generalizability. Section III details the methodology, encompassing the modified PCA formulation and dataset partitioning strategy. Section IV presents the full experimental procedure. Section V reports and analyzes the results, with direct comparison to the prior framework. Section VI contains all references cited in this work.

II. Related Works

A. *Data Dimensional Reduction and Principal Components Analysis [5]*

Salem et al. investigated methodological ways for lowering the complexity of high-dimensional datasets while maintaining the analytical integrity of the original variable space. The authors show that as datasets grow and dimensionality, which is increasingly common in modern scientific and engineering applications, examining inter-variable correlations and visualizing structural relationships becomes increasingly difficult. To address this, the paper introduces Principal Component Analysis as a mathematically rigorous technique for converting a set of correlated variables into a smaller collection of uncorrelated principal components, each sorted by the proportion of total variance it captures. The authors demonstrate how PCA can significantly reduce computational overhead and eliminate redundant feature dimensions while retaining the most analytically significant structure of large datasets using practical examples such as the Iris dataset and Singular Value Decomposition implementations. The paper's fundamental point is that PCA, when used with appropriate pre-processing and component selection criteria, is a theoretically valid and practically effective approach for dimensionality reduction that does not forfeit analytical accuracy.

This work has a direct and significant impact on the current study. The previous solution [4] used a mostly unmodified, off-the-shelf PCA formulation with limited domain-specific customization, resulting in a 22% reconstruction fidelity loss. Salem et al. give methodological support for the proposed improvements, particularly in their handling of normalization processes, covariance structure analysis, and principled component selection using variance retention levels. These considerations inform the three modifications central to this study: extending compression to include a second principal component, applying variance-proportional weighting to each component's contribution, and introducing an

integrated global scaling step applied uniformly during transformation rather than sequentially before or after. The modified PCA formulation developed in this work is intended to capture a greater proportion of meaningful quality-related variation and improve reconstruction agreement beyond the 79% benchmark established in previous work by incorporating domain-specific weighting calibrated to the distributional properties of OFIQ metrics and adopting the rigorous pre-processing practices advocated by Salem et al. In this regard, Salem et al. provide not only theoretical validation of dimensionality reduction as a viable approach, but also a practical methodological reference for developing a more precise, robust, and broadly applicable PCA framework tailored to the specific requirements of biometric facial image quality assessment.

B. Effects of Dataset Features on the Performance of Classification Algorithms [6]

The second foundational study looks at the relationship between intrinsic dataset properties and the stability and predicted accuracy of classification algorithms under various scenarios. Rather than proposing a new classifier, the authors focus on systematically identifying the quantifiable dataset attributes that have the greatest influence on classification success. To do this, the study employs a wide array of datasets that differ in size, feature dimensionality, class distribution balance, inter-feature correlation, redundancy, and noise level. A set of quantitative meta-features is extracted from each dataset, including attributes like the number of features and instances, the attribute-to-instance ratio, the degree of class imbalance, and feature overlap measures, and the performance of multiple classification algorithms is statistically evaluated against them. The data show that high dimensionality relative to sample size, feature redundancy, class imbalance, and overlapping class distributions all have a significant negative impact on classification performance. Conversely, datasets with balanced class distributions, reduced redundancy, and informative feature spaces consistently provide more stable and accurate models. The study concludes that dataset structure has an impact on algorithmic performance that is comparable to, and in some cases greater than, that of algorithm selection itself, emphasizing the critical importance of meticulous pre-processing, feature engineering, and dimensionality reduction as prerequisites for consistent model behavior.

This work has direct and significant implications for the current study, which applies a modified PCA framework to OFIQ-derived facial quality metrics across multiple large-scale datasets — specifically VGGFace2, Flickr Faces HQ, and CelebA — each with distinct structural and distributional properties. These datasets differ in terms of pose variation, acquisition environment, resolution, identity distribution, and demographic makeup, which are the types of structural variations that this study highlights as important for model performance and stability. The extent to which the suggested PCA adjustments preserve reconstruction fidelity across various datasets will thus be influenced not only by the PCA formulation, but also by the statistical traits inherent in each dataset. This finding inspires the cross-dataset validation strategy used in the current study, which involves creating separate training and validation partitions for each dataset and subjecting validation images to artificial luminance and margin modifications to simulate suboptimal real-world capture conditions. This study identified distributional divergence, redundancy, and dimensionality factors as critical determinants of classification robustness. The evaluation framework developed here is designed to provide a rigorous and ecologically valid assessment of the modified PCA's generalizability across diverse biometric imaging environments.

III. Methodology

This study's methodology is structured around three primary objectives: the implementation of a modified PCA formulation designed to improve reconstruction fidelity over the prior framework; the application and validation of that formulation across multiple large-scale and demographically diverse facial image datasets; and the evaluation of the resulting composite quality scores against original OFIQ metrics under both standard and artificially degraded imaging conditions. The subsections that follow describe each component of the pipeline, from dataset gathering and partitioning to PCA training, validation, and performance evaluation.

A. Datasets

This study uses three facial picture datasets, each with its own unique structural and demographic properties. VGGFace2 is a large-scale collection of over 3.3 million photos from over 9,000 identities, with significant variations in pose, age, race, and illumination. Flickr Faces HQ (FFHQ) features 70,000 high-resolution photographs from Flickr, representing a wide range of ages, ethnicities, and image quality. CelebA contains approximately 200,000 celebrity facial photos tagged with 40 binary features, representing a diverse variety of facial looks and capture situations. Together, these datasets provide a breadth of identification, demographic, resolution, and

acquisition environment variety that no single dataset could give on its own, allowing for a more rigorous and ecologically sound examination of the updated framework's generalizability.

Each dataset is partitioned separately between training and validation subsets using an 80/20 split at the individual dataset level. Training sets are only utilized for fitting PCA objects, whereas validation sets are used to assess reconstruction fidelity. Importantly, a separate and independent PCA object is trained for each dataset, rather than fitting a single unified model to all datasets combined. This decision provides several significant advantages. First, it ensures that each PCA item is optimally calibrated to its source dataset's specific distributional qualities, taking into consideration variances in pose variation, resolution, illumination range, and demographic composition that would otherwise be averaged out in a pooled model. Second, it enables a more exact and interpretable assessment of how well the changed formulation generalizes inside each dataset's distinct feature space, rather than aggregating performance across structurally unrelated data. Third, it simulates realistic deployment conditions, in which a quality assessment model is often trained on data from the target acquisition context rather than a generic pooled corpus. The independence of each PCA object improves both the evaluation's internal validity and the conclusions' practical applicability.

B. OFIQ Metric Extraction

After partitioning the dataset, OFIQ is applied to all photos in both the training and validation subsets to extract scalar quality measures. OFIQ is run through its command-line interface, OFIQSampleApp, which generates a structured CSV output file for each image subset, with each row representing a single image and each column representing a unique scalar quality measure. The metric groupings established in previous work remain unchanged, as the brightness, occlusion, and margin categories were obtained via a cross-correlation analysis that revealed consistent pairwise correlations across OFIQ measurements. These groupings serve as the input feature sets for the PCA training technique outlined in the following section.

C. Modified PCA Formulation

The primary contribution of this study is a modified PCA formulation that expands and refines the single-component technique used in previous work. The formulation includes three targeted modifications: the addition of a second principal component, variance-proportional weighting of both components, and an integrated global scaling step that is applied uniformly throughout transformation. Each is detailed in detail below. Before performing PCA fitting, the raw OFIQ metric values within each group are subjected to an integrated global scaling step. Rather than applying normalization before or after the PCA transformation as distinct operations—as was done in previous work—all features are normalized within a single unified phase during transformation using z-score normalization:

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j} \qquad [1]$$

where x_{ij} is the raw value of metric j for image i, μ_j is the mean of metric j across all training images, and σ_j is the corresponding standard deviation. This integrated approach ensures that all features enter the PCA decomposition on a common scale, eliminating the inconsistencies introduced by sequential scaling operations and producing a more stable covariance structure for decomposition. Following global scaling, PCA is applied to each metric group individually. The scaled data matrix is centered and deconstructed using eigen analysis of the covariance matrix C:

$$C = \frac{1}{n-1} Z^T Z \qquad [2]$$

where Z is the matrix of scaled and centered feature values and n is the number of training images. The eigenvalues $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_p$ and their corresponding eigenvectors $v_1, v_2, \ldots, v_p$ are obtained by solving:

$$Cv_k = \lambda_k v_k \qquad [3]$$

Rather than keeping simply the first principal component, as in the previous framework, both PC1 and PC2 are extracted for each group, capturing a larger share of the overall variation in each metric group. The projected scores for each component are calculated as follows:

$$PC1_i = Z_i * v_i \,, PC2_i = Z_i * v_2 \qquad [4]$$

where Z_i is the scaled feature vector for image i and v_1, v_2 are the first and second eigenvectors respectively. The two components are then combined into a single composite quality score using variance-proportional weighting, wherein each component's contribution is scaled by its proportion of total explained variance:

$$w_1 = \frac{\lambda_1}{\lambda_1 + \lambda_2} \,, w_2 = \frac{\lambda_2}{\lambda_1 + \lambda_2} \qquad [5]$$

$$Q_i = w_1 * PC1_i + w_2 * PC2_i \qquad [6]$$

where Q_i is the final composite quality score for image i, and w_1 and w_2 are the variance-proportional weights assigned to PC1 and PC2 respectively. This weighting scheme ensures that the dominant axis of variation retains proportionally greater influence over the composite score, while the secondary component contributes in a statistically grounded manner rather than equally or arbitrarily.

D. Validation Image Augmentation

To assess reconstruction fidelity under settings reflective of real-world suboptimal capture environments, validation set photos are exposed to two types of artificial modification prior to OFIQ metric extraction, as described in the previous work. Luminance modifications are applied by adjusting each image's luminance channel by a randomly sampled factor within the range of ±40%. This produces both darkened and brightened variants to simulate variable illumination conditions encountered in surveillance, outdoor, and non-cooperative biometric acquisition contexts. Margin modifications are made by first detecting the primary face in each image with a bounding box, then computing the distance between each side of the bounding box and the corresponding image edge, and finally applying a randomly sampled crop value within those bounds to each side independently, introducing variability in head positioning and facial framing consistent with irregular camera angles and non-standardized user positioning. All modified photos are stored as independent outputs, while the original validation images are kept for reference. OFIQ is then applied to the modified images to extract the scalar quality metrics needed in reconstruction evaluation.

E. Evaluation Procedure

After extracting OFIQ measures from both unmodified and augmented validation sets, trained PCA objects for each dataset are used to calculate composite quality scores for each image. Performance is assessed by calculating the Pearson correlation coefficient between the PCA-derived composite scores and each of the original constituent OFIQ measures within each group, as well as an overall group-level correlation representing the mean pairwise agreement. These results are directly compared to the preceding work's 79% reconstruction fidelity standard to determine how much the modified formulation improves agreement. Each dataset is evaluated independently, allowing for a cross-dataset comparison of reconstruction stability and insight into how each dataset's structural attributes affect the updated PCA framework's performance.

IV. Experimental Setup

Our first step was to download the images from the three mentioned datasets. For each dataset, given the number we downloaded we decided to maintain as close as an 80/20 split between training and validation data. This is to make sure that we had enough for both functions without over validating and under training. Furthermore, since these databases can encompass hundreds of thousands of images, we decided to significantly reduce the training and validation sizes. Each database has about 1,000 images for training and anywhere from 200 to 350 images for validation. This is due to the computational constraints we have during our study. As seen in Figure 1 below, we have an example of what our training folder contains. Another important note was that the images were taken at random from each dataset.

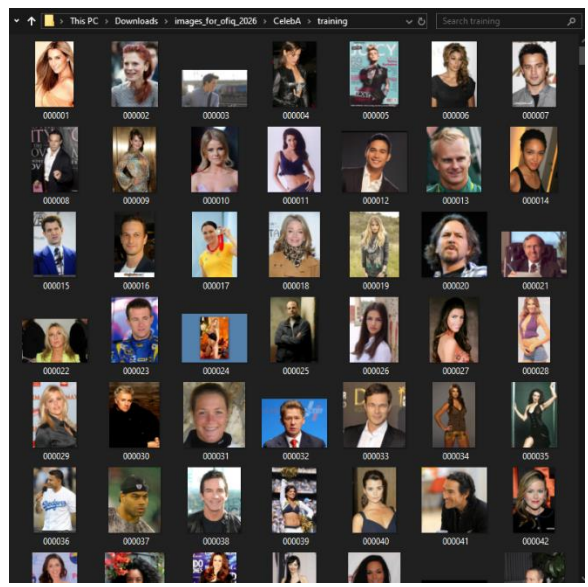

Figure 1: Training Folder for CelebA Dataset

Next, we initialized the OFIQ program. This required installing it from its GitHub and following the steps taken in the previous study. The goal was to get metrics for all three databases' images, and this was simply accomplished by directing the program to the folders with the validation and training data. As seen in Figure 2 below, we have some examples of the OFIQ being used to create the .csv files containing the metrics that we need. An example of one of these .csv files open can be seen in Figure 3 below.

Figure 2: OFIQ Initializing

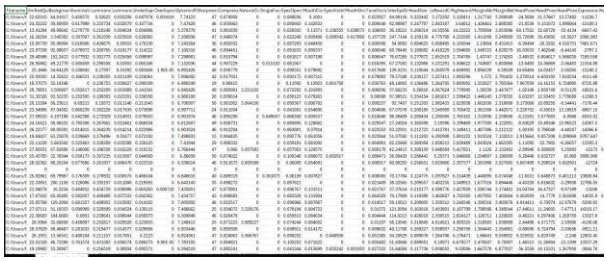

Figure 3: Example of Training Dataset .csv File

Our next step is to retrieve the previous study's code. This was done by referencing the code and re-creating it based on previous descriptions and what we could see from screenshots. After the code was done, we began modifying the PCA function. The first step was to add lines to create and reference a second component to pick up on more of the relationship between the individual components and the created groups. Figure 4 below shows some examples of these lines of code.

```
# --- Changed: n_components=2 to capture more variance ---
pca = PCA(n_components=2)
components = pca.fit_transform(data[cols])

# --- Changed: store pc1 and pc2 as separate columns ---
data[f"{group_name}_pc1"] = components[:, 0]
data[f"{group_name}_pc2"] = components[:, 1]

pca_models[group_name] = pca

# --- Changed: report variance for both components ---
pc1_pct = pca.explained_variance_ratio_[0] * 100
pc2_pct = pca.explained_variance_ratio_[1] * 100
total_pct = pc1_pct + pc2_pct
print(f"Group '{group_name}':")
print(f"  PC1 explains {pc1_pct:.2f}% | PC2 explains {pc2_pct:.2f}% | Total: {total_pct:.2f}%")
```

Figure 4: Code for Multiple PCA Components

Furthermore, Figure 5 below shows how we added a more global scaling to the project. This was implemented by adding an additional PCA model labeled minmax_scaler. The minmax_scaler is a preserved record of the 0-1 scaling boundaries learned from the training data; it remembers the precise minimum and maximum values seen during training, allowing any new data to be scaled to the same range using the same reference points. The old code applies fit_transform() to the test data, which means that a PCA score of 0.8 in training and 0.8 in testing could scale to completely different numbers if the two datasets have different ranges, rendering training and test findings incomparable. The revised code only calls transform(), so the test data is always scaled using the training data's boundaries, ensuring that both sets use the same numerical language.

```
# Scale all 6 PCA columns to 0-1
scaled_cols = [f"{col}_scaled" for col in pc_cols]
scaler = MinMaxScaler(feature_range=(0, 1))
data[scaled_cols] = scaler.fit_transform(data[pc_cols])
```

Figure 5: Minmax_scaler PCA Object Created

Once this is created, there is only one step left to complete before running the code. The step would be to, as in the previous study, to pre-process some of the images randomly with changes in margins and luminosity. We did this by using the following two code blocks, Figures 6 and 7, to randomize these values. The code then saves these images in new folders specifically labeled as "modified_validation_images" to ensure no mix up with other images. An example of the modified image can be seen in Figure 8 below.

```
def randomize_luminance(folder_path, min_factor=0.6, max_factor=1.4, output_suffix='_lum',
                        file_extensions=('jpg', 'jpeg', 'png')):
    """
    Randomly adjusts brightness of every image in folder_path by a factor
    between min_factor and max_factor. A factor below 1.0 darkens the image,
    above 1.0 brightens it.

    Automatically detects whether folder_path is a Google Drive path or a
    local PC path, and saves altered images to the appropriate output location.
    Originals are never overwritten.
    """


    # --- Validate inputs before doing any work ---
    if min_factor < 0 or max_factor < 0 or min_factor > max_factor:
        raise ValueError("Bounds must be non-negative and min_factor must be <= max_factor.")

    # --- Detect path type and mount drive if needed ---
    is_drive_path = detect_and_mount_drive(folder_path)

    if not os.path.exists(folder_path):
        raise FileNotFoundError(f"Folder not found: '{folder_path}'. Check the path and try again.")

    # --- Determine output folder based on path type ---
    output_dir = get_output_folder(folder_path, is_drive_path, output_suffix)

    processed = 0
    skipped = 0
```

Figure 6: Section of Code for Luminance Randomization

```
for fname in os.listdir(folder_path):

    base, ext = os.path.splitext(fname)

    # --- Skip non-image files ---
    if ext.lower() not in file_extensions:
        continue

    # --- Skip already-altered files ---
    if output_suffix in base:
        print(f"Skipping already-altered file: '{fname}'")
        continue

    img_path = os.path.join(folder_path, fname)
    img = cv2.imread(img_path)

    if img is None:
        print(f"Could not read '{fname}', skipping.")
        skipped += 1
        continue

    h, w = img.shape[:2]

    # --- Detect face bounding box ---
    detection = detect_face_opencv(img, face_cascade)
    if detection is None:
        print(f"No face detected in '{fname}', skipping.")
        skipped += 1
        continue

    x_min, y_min, x_max, y_max = detection

    # --- Clamp bounding box to image boundaries ---
    x_min = max(0, x_min)
    y_min = max(0, y_min)
    x_max = min(w, x_max)
    y_max = min(h, y_max)

    # --- Calculate max safe crop from each side ---
    # These are the pixel gaps between the face box and each image edge
    left   = x_min
    top    = y_min
    right  = w - x_max
    bottom = h - y_max

    # --- Randomly crop between 0 and the max safe amount per side ---
    crop_l = random.randint(0, left)
    crop_t = random.randint(0, top)
    crop_r = random.randint(0, right)
    crop_b = random.randint(0, bottom)

    x1, y1 = crop_l, crop_t
    x2, y2 = w - crop_r, h - crop_b
```

Figure 7: Section of Code for Margins Randomization

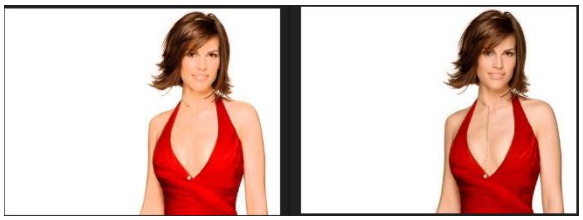

Figure 8: Modified Image (Left) and Original Image (Right)

Once randomly altered data validation images were created for each dataset, we were able to begin the experiment. The process begins with using the code to create grouped metrics based off the OFIQ's original metrics. This can be seen in Figure 9 below as the original csv file now has new columns. Also, note how now we have 6 columns for the 3 groups, with each group getting a PC1 and PC2 value. This also creates the four PCA objects (3 groups + 1 scaler). We then use those PCA objects to generate the group values in the altered validation sets. This can be seen in figure 10 below with the columns on the validation set.

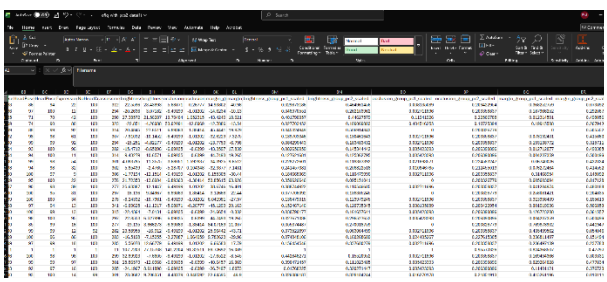

Figure 9: CSV of Processed OFIQ Data

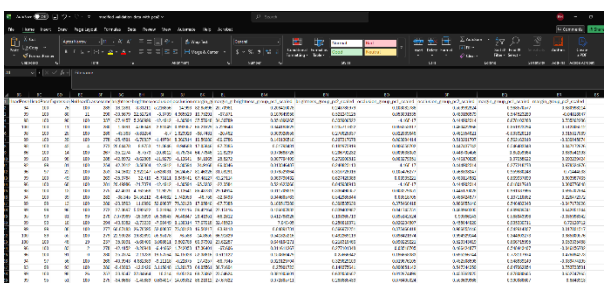

Figure 10: Modified Data Validation CSV File with PCA Object Applied Values

After all of this is considered, we end up with 3 final CSV files for the overall experiment. This would be one per dataset. To process these images, we decided on using basic graphical representation to show the accuracy between the groups and the metrics. We decided to follow the previous study and compare the groups to the individual metrics that make up and see how closely the groups correlate. Given that we now have two PCA objects for each group, we decided to see their individual accuracy, as well as their combined accuracy for the modified validation sets.

V. Results and Analysis

After running the code and comparing the output to the previous study's results, we were able to see significant changes. For example, For the CelebA data of the altered validation images, we saw significant improvements in the Margins and Occlusions groups, as seen in Figures aa and 12 below. This signifies that the additional PCA component was able to cover for some of what the first component, pc1, did not cover.

	Margins
0.76	margins pc1 vs headsize
0.62	margins pc2 vs headsize
0.83	margins pc1+pc2 vs headsize
-0.89	margins pc1 vs MarginAboveOfTheFaceImage
0.09	margins pc2 vs MarginAboveOfTheFaceImage
-0.68	margins pc1+pc2 vs MarginAboveOfTheFaceImage
-0.74	margins pc1 vs MarginBelowOfTheFaceImage
-0.89	margins pc2 vs MarginBelowOfTheFaceImage
-0.92	margins pc1+pc2 vs MarginBelowOfTheFaceImage
81%	Overall correlation of pc1+pc2

Figure 11: Margins Correlation Results of the CelebA Altered Validation Images dataset

	Occlusions
-0.84	occlusions pc1 vs eyesvisable
-0.82	occlusions pc2 vs eyesvisable
-0.91	occlusions pc1+pc2 vs eyesvisable
-0.83	occlusions pc1 vs mouthocclusionprevention
0.17	occlusions pc2 vs mouthocclusionprevention
-0.75	occlusions pc1+pc2 vs mouthocclusionprevention
-0.95	occlusions pc1 vs faceocclusionprevention
-0.44	occlusions pc2 vs faceocclusionprevention
-0.95	occlusions pc1+pc2 vs faceocclusionprevention
87%	Overall correlation of pc1+pc2

Figure 12:Occlusions Correlation Results of the CelebA Altered Validation Images dataset

However, there was a noticeable decrease in the Brightness group from the previous study, as seen in Figure 13 below. However, it should be noted that overall, the brightness group was lower, pointing to this being due to the dataset presenting a challenge to the brightness group PCA. This makes sense, as CelebA would be the most nonstandard and complex

dataset with images from all different types of people and scenarios. To simplify, CelebA presented the most difficult in terms of diversity, and still we were able to see significant growth in accuracy while minor losses were incurred.

Brightness	
-0.99	brightness pc1 vs luminancemean
-0.10	brightness pc2 vs luminancemean
-0.94	brightness pc1+pc2 vs luminancemean
-0.60	brightness pc1 vs luminancevariance
-0.93	brightness pc2 vs luminancevariance
-0.73	brightness pc1+pc2 vs luminancevariance
-0.79	brightness pc1 vs dynamicrange
-0.44	brightness pc2 vs dynamicrange
-0.82	brightness pc1+pc2 vs dynamicrange
83%	Overall correlation of pc1+pc2

Figure 13:Brightness Correlation Results of the CelebA Altered Validation Images dataset

VI. CONCLUSION

This study proposed and evaluated three targeted changes to the PCA-based OFIQ metric compression framework established in previous work: the addition of a second principal component, variance-proportional weighting of PC1 and PC2, and an integrated global scaling step applied uniformly during transformation. Validation was performed on the CelebA dataset using digitally changed photos with tweaked luminance and margin values to replicate suboptimal real-world capture settings. The results showed significant gains in reconstruction fidelity for the occlusion and margin groups, with total correlations of 87% and 81%, respectively, exceeding the 79% threshold established in the previous framework. Taken together, these findings show that adding a second principal component with variance-proportional weighting and integrated global scaling results in a more accurate and generalizable quality compression framework, which is a significant step toward making PCA-based OFIQ compression a more viable and practical tool for large-scale biometric quality assessment.

VII. References

1. Meyer, Charles Piller Josh, and Tom Gorman. "Criminal Faces in the Crowd Still Elude Hidden ID Cameras." *Los Angeles Times*, 2 Feb. 2001, www.latimes.com/archives/la-xpm-2001-feb-02-mn-20035-story.html. Accessed 14 Dec. 2023.
2. O. for, "ISO/IEC 29794-5:2025," *ISO*, 2025. Available: https://www.iso.org/standard/81005.html
3. "Bundesamt für Sicherheit in der Informationstechnik," *Federal Office for Information Security*, Jan. 15, 2025. Available: https://www.bsi.bund.de/EN/Themen/Unternehmen-und-Organisationen/Informationen-und-Empfehlungen/Freie-Software/OFIQ/OFIQ_node.html
4. E. Mendoza, "Redundancy Reduction and Dimensionality Reduction using Principal Component Analysis for Open-Source Face Image Quality Assessment," unpublished.
5. Salem, Nema, and Sahar Hussein. "Data Dimensional Reduction and Principal Components Analysis." Procedia Computer Science, vol. 163, 2019, pp. 292–299, https://doi.org/10.1016/j.procs.2019.12.111.
6. Kwon, Ohbyung, and Jae Mun Sim. "Effects of Data Set Features on the Performances of Classification Algorithms." Expert Systems with Applications, vol. 40, no. 5, Apr. 2013, pp. 1847–1857, https://doi.org/10.1016/j.eswa.2012.09.017.
7.